MIL-D-45472C(MU)
30 APRIL 1965
SUPERSEDING
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8 MARCH 1963

MILITARY SPECIFICATION

DEMOLITION KIT, PROJECTED CHARGE, M157 PARTS, ASSEMBLY AND PACKING

1. SCOPE

1.1 This specification covers the nose section assembly, tail section assembly, body section assembly, center section assembly, and explosive fuze container assembly for one type of mine clearing item designated as Demolition Kit, Projected Charge, M157.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550 — Ammunition and Special Weapons, General Specification for.

MIL-I-45607 —Inspection Equipment, Supply and Maintenance of.

MIL-T-5021 — Tests; Aircraft and Missile Welding, Operator's Qualification.

STANDARDS

MILITARY

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes. (ABC-STD-105).

MIL-STD-109 — Quality Assurance Terms and Defiinitions.

MIL-STD-129 — Marking for Shipment.

MIL-STD-171 — Finishing of Metal and Wood Surfaces.

MIL-STD-1167 — Ammunition Data Cards.

MIL-STD-1235 — Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes.

DRAWINGS

ORDNANCE CORPS

8836900 — Insert Tube Lid.

FSC 1375

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8836906 — Plug Loading.

8836907 — Explosive Fuze Container Assembly.

8837997 — Box, Packing, Ammunition, for Impact Fuze Section Assembly for Demolition Kit, Projected Charge, M157.

8837998 — Box, Packing, Ammunition, for Nose Section Assembly for Demolition Kit, Projected Charge, M157.

8837999 — Box, Packing, Ammunition, for Tail Section Assembly for Demolition Kit, Projected Charge, M157.

8838000 — Box, Packing, Ammunition, for Fuze Container Assembly for Demolition Kit, Projected Charge, M157.

8838001 — Protector End, Packing, Ammunition, for Body Section Assembly or Center Section Assembly for Demolition Kit, Projected Charge, M157.

8838124 — Nose Section Assembly.

8838125 — Impact Fuze Section Assembly.

8838126 — Body Section Assembly.

8838128 —Center Section Assembly.

8838152 — Tail Section Assembly.

PUBLICATIONS

ORDNANCE CORPS

IEL–8836910 — Index of Inspection Equipment Lists.

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. Materials and parts shall be in accordance with applicable drawings and specifications.

3.2 Metal parts. The sub-assemblies shall comply with all requirements specified on drawings (dwgs.) 8838124, 8838125, 8838126, 8838128, 8836907 and 8838152, and with all requirements specified in applicable drawings and specifications.

3.3 Welding. All welds shall be reasonably smooth and free of craters, cracks, porosity, undercut and other surface imperfections. They shall exhibit characteristics of fusion, penetration and soundness of weld deposit representative of good welding practice. They shall be free of irregularities which indicate lack of skill or experience on the part of the operator. Slag and weld splatter shall be completely removed prior to application of protective coating.

3.4 Qualification of welders. Only welders qualified in accordance with specification MIL–T–5021 shall be permitted to weld aluminum components. Welders of steel components need not be qualified but the steel welds shall meet all the requirements of 3.3 of this specification.

3.5 Leakage of insert tube. The insert tube in the center and impact fuze assemblies, prior to the application of internal coating, shall not leak, as evidenced by a drop in pressure, when submitted to an internal air pressure of 2 minus .250 pounds per square

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inch for 30 seconds, minimum.

3.6 Structual strength of center section. There shall be no evidence of loose, missing or shearing of pins, bolts and rivets, cracked or split bodies, no change in cross-sectional contour of the body, weld separation, loose top bearing, or any other defects that could affect the assembling or functioning of the items, after they have been subjected to a load of 13,300 pounds, minimum.

3.7 Impact plate test. There shall be no evidence of a .30 ball ammunition penetrating the impact plate when the projectile strikes entirely within the circumference of the impact plate, but not in the area of the boss, at a distance of approximately 100 feet. The plate shall remain intact but slight cracks on the back side shall be permitted. (see 6.5).

3.8 Deflection. With two center sections joined together by the side joint pins and top bearing bars in the tight position, and with one of the center sections supported in the horizontal position on a level surface, the end of the unsupported center section may not vary more than 3/8 inch above to 3/8 inch below the level surface.

3.9 Embrittlement of springs. Impact fuze springs shall not break extended to the minimum point of permanent distortion when tested as specified in 4.4.8.

3.10 Workmanship. All parts and assemblies shall be fabricated and finished in a thorough workmanlike manner. They shall be free of burrs, chips, sharp edges, unblended radii, surface defects, dirt, grease, molding defects, rust, corrosion products and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. Surface coatings shall be continuous except for a few light scratches. All required markings shall be neat and sharply defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL–STD–109 in order to define terms used herein. The provisions of Specification MIL–A–2550 shall apply.

4.1.1 *Submission of product.* At the time the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

(a) A statement that the lot complies with all quality assurance provisions specified within this specification.

(b) Number of units of product inspected.

(c) Results obtained, by defect code, for all inspection performed.

(d) Drawing, specification number and date, together with an identification and date of changes.

(e) Statement of findings on all material purchased by the contractor when such material is controlled by Government specifications referenced in any of the contractual documents.

(f) Number of items in the lot.

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(g) Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.2 *Government verification.* Using the contractor's written quality assurance procedure, this detail specification, the applicable drawings and other contractual documents as a guide, the Government inspector shall verify at unscheduled intervals all quality assurance operations performed by the contractor. Verification will be performed to the extent necessary to assure compliance with the contractual requirements. Severity of Government inspection of individual characteristics will be directly related to the seriousness of the classification assigned. In no instance will a characteristic classified "critical" be accepted solely on the basis of the contractor's records.

4.2 Initial production inspection.

4.2.1 *Submission.* The contractor, prior to initiation of regular production, shall submit an initial production sample to a Government approved facility as designated by the contracting officer (see 6.4) for evaluation. The initial production sample shall consist of 2 each of the following assemblies unpainted; Nose Section Assembly (dwg. 8838124), Impact Fuze Section Assembly (dwg. 8838125), Center Section Assembly (dwg. 8838128), Tail Section Assembly (dwg. 8838152), Explosive Fuze Container (dwg. 8836907), and 2 complete sets of parts unpainted; of each of the assemblies, except components common to each assembly which have been produced by the contractor or furnished by a supplier and which will be used in fulfilling the contract. All parts and materials, including packaging and packing, shall be obtained from the same sources of supply as will be used in regular production. The sample shall be accompanied by a statement of findings for materials. An initial production sample, or portion thereof, as directed by the contracting officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days, or whenever a change occurs in manufacturing process, material used, drawing, specification, or source of supply as to significantly affect product uniformity as determined by the Government. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample. A sample containing known defects will not be submitted unless specifically authorized by the contracting officer.

4.2.2 *Inspection to be performed.* Assemblies, subassemblies and components will be subjected by the Government to any or all of the examinations and tests specified in this specification and any or all requirements of the applicable drawings.

4.2.3 *Rejection.* If any assembly, subassembly or component fails to comply with any applicable requirement, the initial production sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of an assembly or component in the sample to comply with any of the stated requirements. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit a new initial production sample or portion thereof. Until an initial production sample is accepted, the contractor is, in no way, authorized by the Government to initiate regular production unless otherwise directed by the contracting officer.

4.3 Inspection provisions.

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4.3.1 *Lot formation.* **A lot shall consist of** demolition kit sub-assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification, and same specification revision. Drawing, specification and process changes not affecting safety, performance, or fit, as determined by the Government shall not necessitate changing the lot interfix number.

4.3.2 *Examination.* **Sampling plans and** procedures for the following classifications of defects shall be in accordance with Standard MIL–STD–105 (ABC–STD–105) except that inspection for critical defects, when listed, shall be 100 percent. Continuous sampling plans in accordance with Standard MIL–STD–1235 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to the individual characteristics listed using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 Side joint pin assembly (see dwg. 8837261 covering a detail of dwg. 8838124, 8838125, 8838126 and 8838128).

Categories	*Defects*	*Method of inspection*	*Code No.*
Critical: None defined			
Major:	AQL 0.40 percent		
101.	Balls do not move freely inside joint pin	Manual	01001
102.	360 degree stake, omitted or improper	Visual-Manual	01002
Minor:	AQL 1.5 percent		
201.	Diameter over balls	Gage	01003
202.	Protective coating missing	Visual	01004
203.	Evidence of poor workmanship (see 3.10)	Visual	01005

4.3.2.2 Top bearing bar (see dwg. 8836918 covering a detail of dwgs. 8838124, 8838125, 8838126 and 8838128).

Categories	*Defects*	*Method of inspection*	*Code No.*
Critical: None defined.			
Major:	AQL 0.65 percent		
101.	Location of hole	Gage	02001
102.	Width	Gage	02002
103.	Thickness	Gage	02003
104.	Notch on ends, omitted or mislocated	Gage	02004
Minor:	AQL 2.5 percent		
201.	Distance between recesses	Gage	02005
202.	Width of recess, minimum (min.)	Gage	02006
203.	Distance, end to recess	Gage	02007
204.	Protective coating missing	Visual	02008
205.	Evidence of poor workmanship (see 3.10)	Visual	02009

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4.3.2.3 Pushing bar (see dwg. 8836914 covering a detail of dwgs. 8838124, 8838125, 8838126 and 8838128.

Categories	*Defects*	*Method of inspection*	*Code No.*
Critical: None defined			
Major:	AQL 0.65 percent		
101.	Total length	Gage	03001
102.	Thickness	Gage	03002
103.	Width	Gage	03003
Minor:	AQL 1.50 percent		
201.	Protective coating missing	Visual	03004
202.	Taper missing or inadequate	Visual	03005
203.	Evidence of poor workmanship (see 3.10)	Visual	03006

4.3.2.4 *Link* (see dwg. 8836919 covering a detail of dwgs. 8838124, 8838125, 8838126 and 8838128).

Categories	*Defects*	*Method of inspection*	*Code No.*
Critical: None defined			
Major:	AQL 0.40 percent		
101.	Length inside	Gage	04001
Minor:	AQL 1.50 percent		
201.	Width inside	Gage	04002
202.	Protective coating missing or inadequate	Visual	04003
203.	Evidence of poor workmanship (see 3.10)	Visual	04004

4.3.2.5 Instruction plate (see dwg. 8836083 covering a detail of dwg. 8838124 and 8838152).

Categories	*Defects*	*Method of inspection*	*Code No.*
Critical: None defined			
Major:	AQL 0.40 percent		
101.	Marking missing, misleading or unidentifiable	Visual	05001
Minor: None defined.			

4.3.2.6 Body right and body life, nose section (see dwgs. 8837292 and 8837293 covering details of dwg. 8836905).

Categories	*Defects*	*Method of inspection*	*Code No.*
Critical: None defined.			
Major:	AQL 0.65 percent		
101.	Profile of cross section, min.	Gage	06001
102.	Weld inadequate when applicable (see 3.3)	Visual	06002
103.	Rivet holes in both ends	Visual	06003
104.	Cracks between holes	Visual	06004
Minor:	AQL 1.00 percent		
201.	Total length	Gage	06005
202.	Evidence of poor workmanship (see 3.10)	Visual	06006

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4.3.2.7 Nose hook assembly (see dwg. 8836044 covering a detail of dwg. 8836905).

Categories	*Defects*	*Method of inspection*	*Code No.*
Critical: None defined			
Major:	AQL 0.40 percent		
101.	Weld inadequate (see 3.3)	Visual	07001
Minor:	AQL 0.40 percent		
201.	Evidence of poor workmanship (see 3.10)	Visual	07002

4.3.2.8 Nose section detail assembly (see dwg. 8836905 covering a detail of dwg. 8838124).

Categories	*Defects*	*Method of inspection*	*Code No.*
Critical: None defined			
Major:	AQL 1.00 percent		
101.	Diameter of side joint pin hole	Gage	08001
102.	Location of side joint pin hole from end of body	Gage	08002
103.	Location of center of side joint pin hole to radius in support pin	Gage	08003
104.	Distance between inside surfaces of female lugs	Gage	08004
105.	Symmetry between top joint center support and web section of body	Gage	08005
106.	Symmetry between slots in top joint supports and top joint center support	Gage	08006
107.	Fastener missing or loosely headed	Visual	08007
108.	Weld inadequate (see 3.3)	Visual	08008
109.	Component or operations missing	Visual	08009
Minor:	AQL 0.40 percent		
201.	Evidence of poor workmanship (see 3.10)	Visual	08010

4.3.2.9 Nose section assembly (see dwg. 8838124).

Categories	*Defects*	*Method of inspection*	*Code No.*
Critical: None defined			
Major:	AQL 1.0 percent		
101.	Instruction plate missing	Visual	09001
102.	Top bearing bar missing	Visual	09002
103.	Top joint stud missing	Visual	09003
104.	Side joint pin missing	Visual	09004
105.	Link missing	Visual	09005
106.	Pushing bar missing or bound	Visual-Manual	09006
Minor:	AQL 1.50 percent		
201.	Protective coating damaged or inadequate	Visual	09007
202.	Tapping screw missing	Visual	09008
203.	Evidence of poor workmanship (see 3.10)	Visual	09009

4.3.2.10 Body (see dwg. 8836118 covering a detail of dwgs. 8836909, 8836115 and 8836116).

Categories	*Defects*	*Method of inspection*	*Code No.*
Critical: None defined			
Major:	AQL 0.40 percent		
101.	Profile of cross section, min.	Gage	10001
102.	Weld inadequate (when applicable) (see 3.3)	Visual	10002
Minor:	AQL 1.00 percent		
201.	Total length	Gage	10003
202.	Evidence of poor workmanship (see 3.10)	Visual	10004

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4.3.2.11 Body section detail assembly (see dwg. 8836115 covering a detail of dwg. 8836126).

Categories	*Defects*	*Method of inspection*	*Code No.*
Critical: None defined			
Major:	AQL 1.50 percent		
101.	Diameter of side joint pin hole	Gage	11001
102.	Location of side joint pin hole from end of body	Gage	11002
103.	Location of center of side joint pin hole to radius in support pin	Gage	11003
104.	Distance between inside surfaces of female lugs	Gage	11004
105.	Symmetry between top joint center support and web section of body	Gage	11005
106.	Symmetry between slots in top joint supports and top joint center support	Gage	11006
107.	Distance between surfaces of male lugs	Gage	11007
108.	Disc missing	Visual	11008
109.	Fastener missing or loosely headed	Visual	11009
110.	Weld inadequate (see 3.3)	Visual	11010
111.	Dirt shield missing	Visual	11011
112.	Component or operation missing	Visual	11012
Minor:	AQL 1.00 percent		
201.	Disc color incorrect	Visual	11013
202.	Evidence of poor workmanship (see 3.10)	Visual	11014

4.3.2.12 Insert tube lid (see dwg. 8836900).

Categories	*Defects*	*Method of inspection*	*Code No.*
Critical: None defined			
Major:	AQL 0.40 percent		
101.	Diameter under flange	Gage	12001
Minor:	AQL 1.50 percent		
201.	Height	Gage	12002
202.	Thickness	Gage	12003
203.	Either coating missing or inadequate	Visual	12004
204.	Evidence of poor workmanship (see 3.10)	Visual	12005

4.3.2.13 Insert tube assembly (see dwg. 8836937 covering a detail of dwgs. 8836909 and 8836116).

Categories	*Defects*	*Method of inspection*	*Code No.*
Critical: None defined			
Major:	AQL 0.40 percent		
101.	Weld inadequate (see 3.3)	Visual	13001
102.	Dimple protrudes outward	Visual	13002
Minor:	AQL 0.40 percent		
201.	Evidence of poor workmanship (see 3.10)	Visual	13003

4.3.2.14 Center section detail assembly (see dwg. 8836116 covering a detail of dwg. 8836128).

Categories	*Defects*	*Method of inspection*	*Code No.*
Critical: None defined			
Major:	AQL 1.50 percent		
101.	Diameter of side joint pin hole	Gage	14001

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Categories	Defects	Method of inspection	Code No.
102.	Location of side joint pin hole from end of body	Gage	14002
103.	Location of center of side joint pin hole to radius in support pin	Gage	14003
104.	Distance between inside surfaces of female lugs	Gage	14004
105.	Symmetry between top joint center support and web section of body	Gage	14005
106.	Symmetry between slots in top joint supports and top joint center support	Gage	14006
107.	Distance between surfaces of male lugs	Gage	14007
108.	Diameter of insert tube hole	Gage	14008
109.	Insert tube missing	Visual	14009
110.	Open end of insert tube not at female end of section	Visual	14010
111.	Dimple in end of insert tube protrudes outward	Visual	14011
112.	Weld inadequate (see 3.3)	Visual	14012
113.	Fastener missing or loosely headed	Visual	14013
114.	Dirt shield missing	Visual	14014
115.	Component or operation missing	Visual	14015
Minor:	AQL 1.00 percent		
201.	Insert tube interior coating not continuous	Visual	14016
202.	Evidence of poor workmanship (see 3.10)	Visual	14017

4.3.2.15 Body and center section assemblies (see dwg. 8838126 and 8838128).

Categories	Defects	Method of inspection	Code No.
Critical: None defined			
Major:	AQL 1.0 percent		
101.	Top bearing bar missing	Visual	15001
102.	Side joint pin missing	Visual	15002
103.	Top joint stud missing	Visual	15003
104.	Link missing	Visual	15004
105.	Pushing bar missing or bound	Visual-Manual	15005
Minor:	AQL 1.00 percent		
201.	Protective coating missing or inadequate	Visual	15006
202.	Evidence of poor workmanship (see 3.10)	Visual	15007

4.3.2.16 Impact fuze housing, rear (see dwg. 8836904 covering a detail of dwg. 8836909).

Categories	Defects	Method of inspection	Code No.
Critical: None defined			
Major:	AQL 0.40 percent		
101.	Distance from door end to container plate slot	Gage	16001
Minor:	AQL 0.40 percent		
201.	Evidence of poor workmanship (see 3.10)	Visual	16002

4.3.2.17 Impact plate (see dwg. 8836932 covering a detail of dwg. 8836927).

Categories	Defects	Method of inspection	Code No.
Critical: None defined			
Major:	AQL 0.40 percent		
101.	Thickness through boss	Gage	17001
Minor:	AQL 1.00 percent		
201.	Protective coating missing or inadequate	Visual	17002
202.	Evidence of poor workmanship (see 3.10)	Visual	17003

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4.3.2.18 Spring, impact fuze (see dwg. 8836929 covering a detail of dwg. 8836927).

Categories	*Defects*	*Method of inspection*	*Code No.*
Critical: None defined			
Major:	AQL 0.65 percent		
101.	Outside diameter	Gage	18001
102.	Diameter of wire	Gage	18002
103.	Assembled height under load	Gage	18003
104.	Active coils	Gage	18004
Minor:	AQL 1.5 percent		
201.	Total number coils	Visual	18005
202.	Protective coating missing or inadequate	Visual	18006
203.	Evidence of poor workmanship (see 3.10)	Visual	18007

4.3.2.19 Bullet impact fuze assembly (see dwg. 8836927 covering a detail of dwg. 8836125).

Categories	*Defects*	*Method of inspection*	*Code No.*
Critical: None defined			
Major:	AQL 0.65 percent		
101.	Weld inadequate (see 3.3)	Visual	19001
102.	Component not assembled properly	Visual	19002
103.	Ring missing or not secure	Visual-Manual	19003
Minor:	AQL 1.50 percent		
201.	Weld not coated	Visual	19004
202.	Protective finish missing or inadequate	Visual	19005
203.	Evidence of poor workmanship (see 3.10)	Visual	19006

4.3.2.20 Plug loading (see dwg. 8836906).

Categories	*Defects*	*Method of inspection*	*Code No.*
Critical: None defined			
Major:	AQL 0.40 percent		
101.	Diameter under flange	Gage	20001
Minor:	AQL 1.50 percent		
201.	Height	Gage	20002
202.	Thickness	Gage	20003
203.	Either coating missing or improper	Visual	20004
204.	Evidence of poor workmanship (see 3.10)	Visual	20005

4.3.2.21 Plate, container (see dwg. 8837080 covering a detail of dwg. 8836907).

Categories	*Defects*	*Method of inspection*	*Code No.*
Critical: None defined			
Major:	AQL 0.40 percent		
101.	Diameter of hole	Gage	21001
Minor: None defined			

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4.3.2.22 Cup, fuze (see dwg. 8836935 covering a detail of dwg. 8836907).

Categories	*Defects*	*Method of inspection*	*Code No.*
Critical: None defined			
Major:	AQL 0.40 percent		
101.	Small inside diameter	Gage	22001
102.	Distance bottom to shoulder	Gage	22002
Minor: None defined.			

4.3.2.23 Explosive fuze container, prior to assembling booster plate (see dwg. 8836907).

Categories	*Defects*	*Method of inspection*	*Code No.*
Critical: None defined			
Major:	AQL 0.40 percent		
101.	Weld inadequate (see 3.3)	Visual	23001
Minor: None defined			

4.3.2.24 Explosive fuze container assembly (see dwg. 8836907).

Categories	*Defects*	*Method of inspection*	*Code No.*
Major:	AQL 1.0 percent		
101.	Inside diameter of fuze cup, min.	Gage	24001
102.	Interior coating not continuous	Visual	24002
103.	Fuze cup insecure	Manual	24003
104.	Weld inadequate (see 3.3)	Visual	24004
105.	Bushing inverted or loose	Visual-Manual	24005
Minor:	AQL 1.50 percent		
201.	Distance between sides maximum (max.)	Gage	24006
202.	Protective coating missing or inadequate	Visual	24007
203.	Evidence of poor workmanship (see 3.10)	Visual	24008

4.3.2.25 Impact fuze section detail assembly (see dwg. 8836909 covering a detail of 8836125).

Categories	*Defects*	*Method of inspection*	*Code No.*
Critical: None defined			
Major:	AQL 2.50 percent		
101.	Diameter of side joint pin hole	Gage	25001
102.	Location of side joint pin hole from end of body	Gage	25002
103.	Location of center of side joint pin hole to radius in support pin	Gage	25003
104.	Distance between inside surfaces of female lugs	Gage	25004
105.	Symmetry between top joint center support and web section of body	Gage	25005
106.	Symmetry between slots in top joint supports to top joint center support	Gage	25006
107.	Diameter of insert tube hole	Gage	25007
108.	Distance between surfaces of male lugs	Gage	25008
109.	Distance from end of body to fuze housing door	Gage	25009
110.	Insert tube missing	Visual	25010
111.	Open end of insert tube not at female end of section	Visual	25011
112.	Dimple in end of insert tube protrudes outward	Visual	25012

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Categories	Defects	Method of inspection	Code No.
113.	Cross missing from fuze housing door	Visual	25013
114.	Dirt shield missing	Visual	25014
115.	Weld inadequate (see 3.3)	Visual	25015
116.	Fastener missing or loosely headed	Visual	25016
117.	Cotter pin missing	Visual	25017
118.	Component or operation missing	Visual	25018
119.	Container stop plate missing from fuze housing	Visual	25019
120.	Fuze housing door does not lock, shut or move freely	Manual	25020
Minor:	AQL 1.00 percent		
201.	Insert tube interior coating not continuous	Visual	25021
202.	Evidence of poor workmanship (see 3.10)	Visual	25022

4.3.2.26 Impact fuze section assembly (see dwg. 8838125).

Categories	Defects	Method of inspection	Code No.
Critical: None defined			
Major:	AQL 1.0 percent		
101.	Bullet impact assembly or block missing from fuze housing	Visual	26001
102.	Top bearing bar missing	Visual	26002
103.	Side joint pin missing	Visual	26003
104.	Top joint stud missing	Visual	26004
105.	Link missing	Visual	26005
106.	Pushing bar missing or bound	Visual-Manual	26006
Minor:	AQL 0.40 percent		
201.	Evidence of poor workmanship (see 3.10)	Visual	26007

4.3.2.27 Base, rear hook (see dwg. 8836078 covering a detail of dwg. 8836080).

Categories	Defects	Method of inspection	Code No.
Critical: None defined			
Major:	AQL 0.40 percent		
101.	Total length	Gage	27001
Minor:	AQL 1.00 percent		
201.	Thickness	Gage	27002
202.	Width	Gage	27003

4.3.2.28 Rear hook assembly (see dwg. 8836080 covering a detail of dwg. 8836926).

Categories	Defects	Method of inspection	Code No.
Critical: None defined			
Major:	AQL 0.40 percent		
101.	Weld inadequate (see 3.3)	Visual	28001
Minor:	AQL 1.50 percent		
201.	Location of plate support	Gage	28002
202.	Distance from end of body to end of base	Gage	28003
203.	Evidence of poor workmanship (see 3.10)	Visual	28004

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4.3.2.29 Body left and body right, tail section (see dwg. 8836924 and 8836925 covering details of dwg. 8836926).

Categories	*Defects*	*Method of inspection*	*Code No.*
Critical: None defined			
Major:	AQL 0.65 percent		
101.	Profile of cross section, min.	Gage	29001
102.	Weld inadequate (when applicable) (see 3.3)	Visual	29002
103.	Cracks near holes	Visual	29003
104.	Rivet holes in both ends	Visual	29004
Minor:	AQL 1.50 percent		
201.	Length of cut out section	Gage	29005
202.	Height of cut out section	Gage	29006
203.	Total length	Gage	29007
204.	Evidence of poor workmanship (see 3.10)	Visual	29008

4.3.2.30 Tail section detail assembly (see dwg. 8836926 covering a detail of dwg. 8838152).

Categories	*Defects*	*Method of inspection*	*Code No.*
Critical: None defined			
Major:	AQL 1.50 percent		
101.	Diameter of side joint pin hole	Gage	30001
102.	Location of side joint pin hole from end of body	Gage	30002
103.	Location of center of side joint pin hole to radius in support pin	Gage	30003
104.	Symmetry between top joint center support and web section of body	Gage	30004
105.	Symmetry between slots in top joint supports to top joint center support	Gage	30005
106.	Distance between surfaces of male lugs	Gage	30006
107.	Distance from end of body to end of top plate	Gage	30007
108.	Rear hook assembly bound	Manual	30008
109.	Dirt shield missing	Visual	30009
110.	Weld inadequate (see 3.3)	Visual	30010
111.	Fastener missing or loosely headed	Visual	30011
112.	Component or operation missing	Visual	30012
Minor:	AQL 1.00 percent		
201.	Protective coating missing or inadequate	Visual	30013
202.	Evidence of poor workmanship (see 3.10)	Visual	30014

4.3.2.31 Tail section assembly (see dwg. 8838152).

Categories	*Defects*	*Method of inspection*	*Code No.*
Critical: None defined			
Major:	AQL 1.0 percent		
101.	Instruction plate missing	Visual	31001
102.	Top bearing bar missing	Visual	31002
103.	Top joint stud missing	Visual	31003
104.	End plate missing	Visual	31004
105.	End plate spring missing	Visual	31005
Minor:	AQL 1.00 percent		
201.	Tapping screw missing	Visual	31006
202.	Evidence of poor workmanship (see 3.10)	Visual	31007

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4.3.2.32 Box, packing, nose or tail section, prior to closing (see dwg. 8837998 or 8837999, as applicable).

Categories	*Defects*	*Method of inspection*	*Code No.*
Critical: None defined			
Major:	AQL 0.40 percent		
101.	Packing component missing	Visual	32001
102.	Unit not tightly packed	Manual	32002
Minor: None defined			

4.3.2.33 Box, packing, impact fuze section or explosive fuze container, prior to closing (see dwg. 8837997 or 8838000, as applicable).

Categories	*Defects*	*Method of inspection*	*Code No.*
Critical: None defined			
Major:	AQL 0.65 percent		
101.	Barrier bag missing	Visual	33001
102.	Packing component missing	Visual	33002
103.	Quantity of component in bag improper	Manual	33003
104.	Unit not tightly packed	Manual	33004
Minor: None defined.			

4.3.2.34 Box, packing, sealed (see 8837997, 8837998, 8837999 or 8838000, as applicable).

Categories	*Defects*	*Method of inspection*	*Code No.*
Critical: None defined			
Major:	AQL 1.0 percent		
101.	Contents of box exposed	Visual	34001
102.	FSC number missing or misleading	Visual	34002
103.	Board broken	Visual	34003
104.	Skid missing (where applicable)	Visual	34004
105.	Harware missing, broken or loose	Visual-Manual	34005
106.	Strapping missing, broken or loose	Visual-Manual	34006
Minor:	AQL 1.50 percent		
201.	Marking missing, misleading or unidentifiable	Visual	34007
202.	Strapping out of position	Visual	34008
203.	Car seal missing, unsealed or improperly positioned	Visual	34009

4.3.2.35 End protector, center section, prior to sealing (see dwg. 8838001).

Categories	*Defects*	*Method of inspection*	*Code No.*
Critical: None defined			
Major:	AQL 0.40 percent		
101.	Barrier bag missing	Visual	35001
102.	Quantity of component in bag improper	Manual	35002
Minor: None defined			

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4.3.2.36 End protector, center or body section, as applicable, after sealing (see dwg. 8838001).

Categories	*Defects*	*Method of inspection*	*Code No.*
Critical: None defined			
Major:	AQL 0.65 percent		
101.	FSC number missing or misleading	Visual	36001
102.	Board broken	Visual	36002
103.	Protector loose	Manual	36003
104.	Strapping missing, broken or loose	Visual-Manual	36004
Minor:	AQL 0.40 percent		
201.	Marking, misleading or unidentifiable	Visual	36005

4.3.3 *Testing.*

4.3.3.1 Initial production sample (see 4.2.1). The samples submitted as the initial production quantity shall be tested as specified below after having been examined for all other drawing requirements. Rejection and resubmission shall be in accordance with 4.2.3.

4.3.3.1.1 Load at assembled height of fuze impact spring (see dwg. 8836629 and 4.4.1) — Code No. 18008.

4.3.3.1.2 Qualification of luminous paint (see dwg. 8836930 and 4.4.2) — Code No. 37001.

4.3.3.1.3 Hardness of impact plate (see dwg. 8836932, MIL–S–11356 and 4.4.3) — Code No. 17004.

4.3.3.1.4 Leakage of insert tube (see 3.5 and 4.4.4) — Code No. 14018.

4.3.3.1.5 Structural strength of center sections (see 3.6 and 4.4.5) — Code No. 15008.

4.3.3.1.6 Ballistic test of impact plate (see 3.7 and 4.4.6) — Code No. 17005.

4.3.3.1.7 Deflection (see 3.8 and 4.4.7) — Code No. 14019.

4.3.3.1.8 Embrittlement of springs (see 3.9 and 4.4.8) — Code No. 18009.

4.3.3.2 Production testing.

4.3.3.2.1 Load at assembled height of fuze impact spring (see dwg. 8836629) — Major defect — Code No. 18008. This determination shall be performed 100 percent at the prime contractors plant using the applicable test equipment of 4.3.4 as specified in 4.4.1. Any unit failing to meet the requirement shall be classed defective and removed from the lot.

4.3.3.2.2 Qualification of luminous paint (see dwg. 8836930) Major defect Code No. 37001. In the event the paint qualified under 4.3.3.1.2 is sufficient for a lot, this test need not be repeated. Any additional paint received or to be used on subsequent lots shall be tested as specified in 4.4.2. Failure to meet the requirement shall reject the paint represented by the sample.

4.3.3.2.3 Hardness of impact plate (see dwg. 8836932 and MIL–S–11356) — Major defect — Code No. 17004. This test shall be performed 100 percent using the applicable test equipment of 4.3.4 as specified in 4.4.3. Any unit exhibiting a hardness outside the permitted range shall be classed defective and removed from the lot.

4.3.3.2.4 Leakage of insert tube (see 3.5) — Major defect — Code No. 14018. This

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test shall be performed 100 percent. Any assembly containing an insert tube exhibiting a pressure drop shall be classed defective and removed from the lot subject to repair and resubmission to test. Testing shall be as specified in 4.4.4 using the applicable equipment of 4.3.4.

4.3.3.2.5 Structural strength of center section (see 3.6) — Major defect — Code No. 15008. A sample of 20 center sections from the first production lot and 10 sections from each lot thereafter shall be submitted to this test. Failure of one or more sections in fully complying with the requirement shall reject the lot. Testing shall be specified in 4.4.5 using the applicable equipment of 4.3.4.

4.3.3.2.6 Ballistic test of impact plate (see 3.7) — Major defect — Code No. 17005. A sample of 10 impact plates from the first heat treatment batch, and 5 plates from each subsequent batch, shall be submitted to this test. The samples shall be selected after hardness testing. If one or more units of a sample fail to meet the requirement, the product represented by that sample shall be rejected subject to reheat treatment, hardness testing and resubmission to test. Testing shall be as specified in 4.4.6.

4.3.3.2.7 Deflection (see 3.8) — Major defect — Code No. 14019. A sample of 20 center sections from the first production lot, and 10 sections from each lot produced thereafter, shall be submitted to this test. Failure of one or more of the testing assemblies in complying with the requirement shall reject the lot. Testing shall be as specified in 4.4.7 using the applicable equipment of 4.3.4.

4.3.3.2.8 Spring embrittlement (see 3.9 and 4.4.8) Major defect Code No. 18009. A random sample of five (5) springs shall be selected from each embrittlement relief batch for these tests. The spring loads shall be determined prior to embrittlement testing. Failure of one or more of the requirements shall reject the batch. Testing shall be performed as specified in 4.4.8 and 4.4.1.

4.3.4 *Inspection Equipment.* Index of Inspection Equipment Number IEL–8836910 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in paragraph 6.3.

4.3.4.1 *Government rights to documentation.* Inspection equipment drawings and lists provided and revised in accordance with the requirements of the IEL may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

4.3.4.2 *Supply and maintenance.* Supply and maintenance of the equipment listed on the IEL shall be in accordance with Specification MIL–I–45607.

4.3.4.3 *Government use of contractor's inspection and test equipment.* The contractor shall make available to the Government all inspection and test equipment necessary for determining conformance with contract requirements. Personnel for operating the equipment, and verification of its accuracy, shall be supplied by the contractor for the performance of examination or test by the Government.

4.4 Test Methods and Procedures

4.4.1 *Spring load.* The spring shall be assembled into the applicable test equipment of 4.3.4 and force applied until the spring is compressed to the larger specified height. The applied force shall be recorded. Force shall be re-applied until the unit is compres-

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sed to the smaller height and the applied force recorded. Any spring exhibiting a compression force, at either height, outside the permitted range, shall be classed defective.

4.4.2 *Qualification of luminous paint.* A cross of the approximate dimensions of dwg. 8836930 shall be painted on a fuze housing door, dwg. 8836979, or on a piece of sheet metal. The painted surface shall be exposed to daylight for the specified period. It shall then be stored in total darkness for the specified period and observation for visibility made from the required distance. If the cross is not discernible, the presented product shall be rejected.

4.4.3 *Hardness of impact plate.* The Rockwell C hardness of the plate shall be determined by use of the applicable test equipment. Any plate exhibiting a hardness outside the permitted range shall be classed defective.

4.4.4 *Leakage of insert tube.* The applicable test equipment shall be assembled to the insert tube of the assembly. (The pressure gauge used shall have graduations of 0.1 pound, minimum, and be accurate to those graduations.) The required internal pressure shall be applied and the air supply disconnected. Observation shall be made during the required period for a pressure drop Evidence of a pressure drop shall class an assembly defective.

4.4.5 *Structural strength of center sections.* Two center sections shall be assembled one to the other. The top bearing bars shall be in the tight position. The pushing bar shall be inserted into the coupled sections with equal length in each section. The coupled sections, inverted from the normal position, shall then be placed across two supports with each support 30 plus or minus .250 inch from the junction of the sections. The specified force shall be applied vertically, forcing the test sections against the supports, and the dwell time shall be 5 seconds, minimum. The force shall be applied equally to each section at their junction and within 2 inches from the end of each section. After test, the sections shall be observed for evidence of structural failure, shall then be disassembled and the cross sectional contour checked. Any test specimen exhibiting structural failure or a change in contour shall be classed defective. Acceptable center sections may be returned to the lot.

4.4.6 *Ballistic test of impact plate.* The impact plate shall be positioned vertically with the flat surface toward the range and shall be backed up by sandbags. The plate shall be hit by the projectile from the specified range. Hits in the boss area shall be disregarded. After a hit in the required section observation shall be made for evidence of penetration greater than permitted. Any plate exhibiting excessive penetration shall be classed defective.

4.4.7 *Deflection.* Two center sections shall be assembled one to the other as specified in 4.4 5 except that the push bar need not be assembled. One section shall be supported horizontally on a level surface and the other shall extend into the air. Measurement shall be taken from the bottom of the center section at the edge of the supporting surface to the bottom edge at the end of the unsupported section. If the difference between the two measurements exceeds the permitted variation, the test assembly shall be classed defective.

4.4.8 *Spring embrittlement.* The springs shall be pulled to the minimum point of permanent distortion by any method approved by the procuring activity. Springs so tested shall be scrapped. If spring breaks during test, it shall be classed defective.

5. PREPARATION FOR DELIVERY

5.1 **Level A and C.** Level A and C packing

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shall be in accordance with 8837997, 8837998, 8837999, 8838000 and 8838001, as applicable.

5.1.1 *Marking.* **Packing cases containing** body, tail and nose sections shall be marked in full accordance with the applicable drawing and destination marking on all containers shall be in accordance with Standard MIL–STD–129, on labels.

5.2 Data cards. Data card information shall be prepared for each lot in accordance with the procedures specified in Specification MIL–STD–1167.

6. NOTES

6.1 Ordering data. Procurement documents shall specify title, number and date of this specification.

6.2 Inspection code numbers. The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.3 Inspection equipment. The contractor shall design inspection equipment as required by the referenced Inspection Equipment Lists (IEL) in accordance with the instructions of paragraphs 6.3.1 through 6.3.7.

6.3.1 *Inspection equipment lists.* Inspection equipment lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (00 Form 1242–3) the numbers of drawings of existing equipment designs or codes as indicated in paragraph 6.3.2. Design action required of the contractor with respect to the different types of drawings that may be listed is described in paragraphs 6.3.3 and 6.3.4. Action required by the contractor with respect to commercial inspection equipment is described in 6.3.5. The contractor will be required to prepare detailed drawings in accordance with 6.3.6 for all the equipment coded as "Contractor Design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

6.3.2 *Inspection equipment list codes.* The inspection equipment as defined in 6.3.3, 6.3.4, 6.3.5 and 6.3.6 will be designed in the IEL by the following codes:

CDAF	— Contractor's design responsibility on Army format in accordance with MIL–D–45608.
CDCF	— Contractor's design responsibility on contractor format.
AD	— Army design.
ADMU	— Army design, mandatory for use.
CE	— Commercial equipment.
SCD	— Specification control drawing.

6.3.3 *Army designs.* **Army designs are reflected** on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications, mandatory and non-mandatory. When an inspection equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs and with the approval of the Government, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

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6.3.4 *Specification control drawing.* Specification control drawings depict the minimum equipment requirements in outline, descriptive, diagramatic or pictorial form only and specify the required performance or other characteristics. Contractors must prepare detailed drawings (see 6.3.6) of their designs in support of specification control drawings. These contractor prepared designs must be approved by the Government prior to the fabrication or procuring of the equipment. Commercial equipment meeting the requirements of specification control drawings may be approved if described in sufficient detail to permit identification and evaluation by the Government. Designs shall be submitted for approval as specified in 6.3.7.

6.3.5 *Commercial Equipment.* Commercial equipment is inspection equipment that has universal application for a specific function. It is comprised of items commonly used by industry and Government. Contractors are not required to furnish drawings of commercial inspection equipment, but a list of such equipment must be approved by the Government. Lists shall be submitted for approval to the inspection element of the agency administering the contract.

6.3.6 *Contractor designs.* Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Contractor designs identified as CDCF may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above. Contractor designs identified as CDAF shall comply with the format and requirements of MIL–D–45608, and in addition, contain the detail and information specified above. Final approved contractor designs identified as CDCF shall be suitable for microfilming. They must be non-reproducible diazo or electrostatic prints of top quality full size with black lines on white background and shall provide a legible print produced via the 35mm microfilm process. The contractor shall submit one rolled or flat set copy of the final approved design.

6.3.7 *Submission of contractor designs.* Designs shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA–ND. Design review will normally be accomplished within one month after receipt by Picatinny Arsenal. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs.

6.4 Submission of initial production samples. Instructions, as to the location for the evaluation of the initial production sample shall be obtained from the contracting officer. Upon receipt of such request, the contracting officer shall advise Picatinny Arsenal and instructions will be issued accordingly. All inquires should be forwarded to: Commanding Officer, Picatinny Arsenal, ATTN: SMUPA–NI.

Custodian:

Army—MU

Preparing activity:

Army—MU

Project No. 1375–A900

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SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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DEPARTMENT OF THE ARMY
Picatinny Arsenal
Dover, New Jersey 07801

POSTAGE AND FEES PAID
DEPARTMENT OF THE ARMY

OFFICIAL BUSINESS

Commanding Officer
Picatinny Arsenal
ATTN: SMUPA-VP
Dover, New Jersey 07801

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